TABLE I
TASK, PLANNING, AND EVALUATION SETTINGS

Task and environment	
Team	Three Crazyflie 2 quadrotors; one load; 0.6 m tension-only cables
Load mass	10–40 g per layout, hidden; nominal mass 25 g
Obstacles	Four pillars, with or without two moving spheres (0.25–0.45 m/s)
Mission	Goal 3.2 m ahead; time limit 20 s
Success criterion	Load within 0.15 m of the goal, below 0.2 m/s, continuously for 0.5 s
Control and planning	
Simulation/control	MuJoCo and onboard Lee controllers, both at 500 Hz
Decision rate	10 Hz
Observation delay	0, 50, 100, or 200 ms
Shared planner	MPPI; 512 candidate plans per decision
Planning horizon	15 decisions (1.5 s)
Setpoint moves	At most 0.12 m per axis
Data and evaluation	
Training budgets	500 / 2,000 episodes; 500 uses the first quarter of the same dataset
Full training set	170,304 decisions (4.7 h); 100 separate validation episodes
Main model set	5 model rows $\times$ 2 budgets $\times$ 3 new seeds $=$ 30 models; 1.28–1.37 M parameters
Test coverage	50 layouts $\times$ 2 obstacle settings $\times$ 4 delays $=$ 400 episodes per model
Probe states	Every fifth decision of U's test flights; 8,660 states in total
Uncertainty	95% intervals; 10,000 bootstrap resamples of the 50 test layouts